

A rank-one working model of behavior leakage

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1 Setup

We study *leakage*: a behavior trained into a model under one context (a system prompt, persona, or ICL prefix) also appearing under other, untrained contexts. The measured quantity is the trained-base log-probability of the behavior token(s) at the end of a bystander context’s own on-policy response.

Represent objects as residual-stream activation vectors at a fixed layer:

- a **context** c by $v_c \in \mathbb{R}^d$, read at the last prompt token (mean over a probe set of user questions);
- a **behavior** b by $v_b \in \mathbb{R}^d$, read at the position where the behavior is emitted (for the marker, the end-of-response slot).

2 What the literature establishes

Five premises, each independently supported.

L1 (weights read and write linear features).

The residual stream is a linear communication channel: every weight matrix interacts with it by *reading* a set of input directions and *writing* a set of output directions [1]. A rank-one matrix $uw^\top$ is the atomic read/write unit: it reads feature w (a dot product) and writes feature u (a scaled addition). LoRA makes this factorization literal: in $\Delta W = BA$, the rows of A are read directions (“if” filters) and the columns of B are write directions; a rank-1 adapter projects the activation onto its A vector to produce a scalar gate on its B vector, which acts as a steering vector on the residual stream [2].

L2 (behaviors are linear write features).

Adding a single direction to the residual stream elicits a behavior (steering vectors [3, 4]); persona traits are readable as single directions at the last prompt token, with projection predicting trait expression before generation [5]; a single “toxic persona” direction most strongly predicts and controls emergent misalignment [6]; independently trained misaligned fine-tunes converge to one shared linear misalignment direction, and EM can be induced by as few as nine rank-1 adapters [7].

L3 (narrow fine-tunes act like constant steering writes, gated by pre-existing reads).

Narrow LoRA fine-tunes “essentially add a constant steering vector”: the added vectors have pairwise cosines near $|1|$ across prompts and tokens, and a trigger-gated (conditional) behavior is reproduced by an *unconditional* constant write, with the gating attributed to pre-existing QK attention circuitry whose job is to read the trigger [8]. Narrow fine-tuning also leaves a context-independent activation bias on entirely unrelated text, whose rank-1 projection is

causally load-bearing [9]. Conditionality, in this picture, is magnitude along a fixed write direction, not a new gated circuit.

L4 (edits are rank-one key→value associations).

Linear associative memories store input–output pairs as outer products, and interference between stored pairs is governed by key overlap [10, 11]. Model editing operationalizes this on transformers: ROME inserts a fact by exactly the minimal rank-one update satisfying $W'k_* = v_*$ (up to key-covariance whitening) [12], and weight-space task vectors compose approximately additively [13].

L5 (fine-tuning is approximately kernel regression).

For small parameter updates, the function change at input x from training on x' is $\Delta f(x) \approx K(x, x') \times (\text{residual at } x')$, where K is the empirical neural tangent kernel [14]; kernel dynamics have been shown to approximate language-model fine-tuning in several regimes [15].

3 From premises to the formula

The map. By L1–L2, take the context to be a read feature (v_c) and the behavior a write feature (v_b), and approximate the model’s pre-existing context-to-behavior wiring at one layer as a linear map,

$$v_b \approx Mv_c. \quad (1)$$

The edit. Suppose context c' originally produces behavior b' ($v_{b'} = Mv_{c'}$) and we fine-tune the model to instead produce b'' under c' . By L3, the fine-tune contributes a near-constant steering write gated by a read of the trained context; by L4, the natural form of such an edit is a rank-one key→value association with key $v_{c'}$ and value $v_{b''} - v_{b'}$. The *minimal* such edit, $\min \|\Delta M\|_F$ subject to $(M + \Delta M)v_{c'} = v_{b''}$, is

$$M' = M + \frac{(v_{b''} - v_{b'})v_{c'}^\top}{\|v_{c'}\|^2}. \quad (2)$$

Three independent routes give the same shape: (i) the constrained least-norm edit above, which is ROME’s update with whitened keys [12]; (ii) one delta-rule gradient step on $\frac{1}{2}\|v_{b''} - Mv_{c'}\|^2$ at the exact-fit step size, which is the classical outer-product update of associative memories [10, 11]; (iii) the first-order kernel expansion of fine-tuning (L5) with the activation inner product standing in for K and $(v_{b''} - v_{b'})$ as the residual.

The consequence. The induced change at any *bystander* context c is

$$\Delta v_b(c) = \underbrace{(v_{b''} - v_{b'})}_{\text{write: what leaks}} \cdot \underbrace{\frac{\langle v_{c'}, v_c \rangle}{\|v_{c'}\|^2}}_{\text{read: who leaks}}. \quad (3)$$

The read/write factorization splits leakage into two separately testable parts: the *read* side (overlap with the trained context’s key) determines which bystanders are affected and by how much; the *write* side (the behavior delta) determines what change they all receive, and is the same direction for everyone. Measured quantities are linear reads of the write feature: at the read-out, $\Delta z_{\text{marker}} = \langle W_U[\text{marker}], \Delta v_b \rangle$, so log-probabilities inherit equation (3) up to softmax compression.

Multiple pairs. Training on several (context, behavior) pairs sums their rank-one terms (L4). This is why contrastive negatives matter: persona contexts are mutually similar (read overlaps near one another), so a positive-only implant writes nearly uniformly to all of them; negative pairs add *negative* outer products that subtract this common mode, exposing the graded read-overlap term as a measurable leakage gradient.

4 What it implies

Each structural feature of equation (3) is a falsifiable prediction.

P0 (the adapter itself decomposes into read/write).

The trained update should be inspectable directly: per layer, the top right-singular vector of ΔW should align with the trained context’s representation (read) and the top left-singular vector with the behavior delta (write) [2]. Testable on stored adapters with no new training.

P1 (shared write direction).

The activation shift at the behavior read-out points in the *same* direction ($v_{b''} - v_{b'}$) for every bystander context; only the scalar coefficient varies. Operationally: stack per-bystander shift vectors, run an SVD, and the top component should dominate and match the pre-training behavior delta.

P2 (read-overlap coefficient).

The per-bystander coefficient is $\langle v_{c'}, v_c \rangle / \|v_{c'}\|^2$. This explains why context cosine similarity predicts leakage, and it is *asymmetric*: the coefficient from $c' \rightarrow c$ is $\cos \theta \cdot \|v_c\| / \|v_{c'}\|$, not invariant under swapping source and bystander. The norms carry the asymmetry.

P3 (residual scaling; behavior dependence).

The update magnitude scales with $\|v_{b''} - v_{b'}\|$, the gap between the trained behavior and what the source context already produces. If the source context already elicits the behavior (e.g. a system prompt instructing it), the residual is near zero and the rule predicts little leakage even when context similarity is high; a similarity-only predictor gets this case wrong. On the read-out side, the softmax compresses $\Delta \log P$ near saturation, so prior-dependence should appear in log-probability space and shrink in logit space.

P4 (additivity).

Training on two pairs sums their rank-one updates, so leakage from joint training should approximately equal the sum of the single-source leakages (the activation-space analogue of task arithmetic [13]).

P5 (layer locality).

The relevant v_c is the one feeding the behavior’s computation, so the most predictive layer should sit where the behavior is computed: late for a next-token marker, earlier for higher-level behaviors such as sycophancy.

P6 (cross-behavior transfer).

The same update moves a *different* behavior b at a fixed context in proportion to the projection of $(v_{b''} - v_{b'})$ onto b ’s read-out direction, extending the rule from context generalization to behavior generalization.

5 How we are testing it

	Test	Status / result
P0	SVD of stored marker/EM adapters; compare top singular directions to $v_{c'}$ and to the measured behavior delta. External replication target: the per-trigger backdoor steering vectors in [8]’s repo, trained with the same behavior under 3–4 triggers but never cross-compared	Not yet run; free on artifacts we already have.
P1	SVD of per-context activation shifts after a marker implant vs misalignment SFT (#521, controls #551)	Surprising split: misalignment SFT shifts all 14 probed contexts along one shared direction ($\cos 0.96\text{--}0.98$, $\sim$ half the spectrum; seed-stable) — consistent with the convergent EM direction of [7]; the marker implant does not, and the trained context is the <i>least</i> aligned. Contrast survives controls. Benign SFT also yields one shared direction (#552, running), so the geometry may be generic to plain SFT; a positive-only marker retrain (#561, running) tests whether the contrastive rig, not the behavior, sets the geometry.
P2	Correlate base-model context distances with measured leakage across source $\times$ bystander panels (#502, #489; sycophancy analogue)	Context similarity ranks marker leakage well ($ \rho \approx 0.76\text{--}0.79$ at layers 19–24) and extends to ICL/multi-turn contexts; on held-out persona panels geometry predicts the training-induced <i>change</i> but the structure does not transfer across panels (#553), and the base-model prior beats geometric distance on instruction contexts (#532). The magnitude half of P2 is unsettled in #521.
P3	Vary how strongly the source context already elicits the behavior; instructed-marker break case; track logit vs log-prob spaces (#500, #531, #532)	Growing support: the bystander’s own prior on the behavior is the most consistently supported predictor across panels; logit-space tracking is now in place to separate read-out compression from the update itself.
P4	Joint two-source training vs sum of singletons (#520, #527, #538)	Not yet diagnostic: additivity cosine reads 0.99, but each singleton’s shift matrix is itself near rank-one, so near-parallel constants sum trivially; the confound gates fail at every tested training depth. A regime where singleton shifts vary across contexts is needed.
P5	Layer sweep of predictor quality per behavior	Consistent: marker leakage is best predicted at layers 19–24; sycophancy at layers 1–8 (best cell: MMD at layer 7).
P6	Behavior-generalization testbed ($B \rightarrow B'$ leakage matrix, #545); context testbed (#537); formula-as-predictor (#510)	Queued: #537 running, #545 and #510 not yet run.

6 Known gaps

- The linear premise (1) is assumed, not fitted. A direct check is to regress v_b on v_c across many context-behavior pairs before training and measure how much variance a single linear map explains.
- The rule lives at one layer; training updates all layers. P5 is a heuristic patch, not a derivation.
- The P1 results carry a training-rig confound (contrastive marker-only loss vs plain SFT) until #561 lands.
- The read-out is nonlinear (softmax over the unembedding), so activation-space predictions must be compared against logits, with log-probabilities treated as a compressed view near saturation.
- The strongest constant-steering evidence ([8, 9]) comes from one research orbit and none of the published regimes is persona-gated LoRA on Qwen-2.5-7B; and the published bridge from LoRA-induced trait shifts to persona directions did not survive our adversarial verification (only the prompt-elicitation projection result in [5] stands).
- Whether training *strength* raises the effective rank or context-dependence of the edit is unmeasured in the literature (the only published rank-raising lever is training-data diversity [9]); #538 found the geometry unchanged at a $3\times$ deeper implant.

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